

Leading-term extraction from the Taylor tree (equal starting exponents)

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Abstract

Groundwork for Astra #14 rank 2 (the unequal-leading-term posterior ratio). For equal starting exponents $p = (h_1 + 1)/k_1 = (h_2 + 1)/k_2$ the retained exponent set of the Taylor tree below $2T \leq p + \min(1/k_1, 1/k_2)$ is $\{p\}$ (`polesBelowGen_eq_singleton`), so the chart integral satisfies $Z(N) = N^{-p}(A_p \log N + B_p) + O(N^{-(p+\delta)}(1 + \log N))$ with $\delta = \min(1/k_1, 1/k_2)/2$ (`chart_leading_isBig0`). Hence $Z \sim A_p N^{-p} \log N$ when $A_p \neq 0$ (`chart_isEquivalent_log`) and $Z \sim B_p N^{-p}$ when $A_p = 0 \neq B_p$ (`chart_isEquivalent_const`): the two leading behaviours whose quotient the calculus of unit 166 handles. Zero sorry/axiom.

1 The things formalised

```
theorem polesBelowGen_eq_singleton (h h k k) (hk hk) (p T) (hp hp) (hT : p < 2 * T)
  (hT : 2 * T - p + 1 / k) (hT : 2 * T - p + 1 / k) :
  polesBelowGen h h k k p p T = {p}
def leadingGap (k k : ) : := min (1 / k) (1 / k) / 2
theorem chart_leading_isBig0 ... :
  (fun N => twoDAmp b N h h k k (anaAmp (ampCoeff x y) b)
    - N ^ (-p) * (canonA ... p * Real.log N + canonB ... p))
    =0[atTop] fun N => N ^ (-(p + leadingGap k k)) * (1 + Real.log N)
theorem isLittle0_remainder_log (p ) (h) (A) (hA : A 0) :
  (fun N => N ^ (-(p + )) * (1 + log N)) =o[atTop] powLog A p 1
theorem isLittle0_const_term (p A B) (hA : A 0) : (fun N => B * N ^ (-
p)) =o[atTop] powLog A p 1
theorem isLittle0_remainder_const (p ) (h) (B) (hB : B 0) :
  (fun N => N ^ (-(p + )) * (1 + log N)) =o[atTop] powLog B p 0
theorem chart_isEquivalent_log ... (hA : canonA ... p 0) :
  (fun N => twoDAmp ...) ~[atTop] powLog (canonA ... p) p 1
theorem chart_isEquivalent_const ... (hA : canonA ... p = 0) (hB : canonB ... p 0) :
  (fun N => twoDAmp ...) ~[atTop] powLog (canonB ... p) p 0
```

2 Mathematics

The exponents of $\Lambda(h, k)$ above the common start p are $p+i/k_1$ and $p+j/k_2$ with $i, j \geq 1$, so nothing lies in $(p, p + \min(1/k_1, 1/k_2))$; taking $2T = p + \delta$ with δ half the gap retains only p and gives the remainder $O(N^{-(p+\delta)}(1 + \log N))$ from the general theorem. The remainder and the constant term are $o(N^{-p} \log N)$, and the remainder is $o(N^{-p})$, by the elementary limits $N^{-\delta}(1 + \log N)/\log N \rightarrow 0$, $1/\log N \rightarrow 0$ and $N^{-\delta}(1 + \log N) \rightarrow 0$.

3 Paper-facing meaning

With unit 166 this gives the deterministic leading posterior ratio for equal starting exponents in all three regimes: $A_p^\varphi \neq 0$ (constant limit $A_p^\varphi/A_p^1 = y_{\varphi,00}/y_{1,00}$), $A_p^\varphi = 0 \neq B_p^\varphi$ (decay like $1/\log N$, the corner-vanishing case Astra #14 highlighted), and the power-decay case, which cannot occur at equal starts within this two-term expansion (it requires the next exponent). The chart-level corollary is the next unit.

4 Lean notes

- `rintro rfl on a goal = p` substitutes the theorem variable `p` away (“unknown identifier `p`”); use `intro h ; rw [h]`.
- `Real.tendsto_log_atTop.inv_tendsto_atTop` produces the Π inverse $\log^{-1}$; rewrite `Pi.inv_apply` before `field_simp`.
- `isLittle0_iff_tendsto'` needs the eventual implication “denominator zero $\rightarrow$ numerator zero”, discharged by contradiction since the model term is nonzero for $N > 1$.
- `Finset.sum_singleton` after rewriting the retained set to `{p}`; `IsBig0.trans_isLittle0` and `IsLittle0.add` assemble the equivalences via `IsLittle0.isEquivalent`.